

A diagram about funcoids and reloids

OPG-60024

L1 - TRACTABLE

PARTIALLY SOLVED

Topology

ORIGINAL PROBLEM STATEMENT

Define for posets with order $\sqsubseteq$:

$$- \Phi_* f = \lambda b \in \mathfrak{B} : \bigcup \{x \in \mathfrak{A} \mid fx \sqsubseteq b\};$$

$$- \Phi^* f = \lambda b \in \mathfrak{A} : \bigcap \{x \in \mathfrak{B} \mid fx \sqsupseteq b\}.$$

Note that the above is a generalization of monotone Galois connections (with max and min replaced with suprema and infima).

Then we have the following diagram:

What is at the node "other" in the diagram is unknown.

Conjecture "Other" is $\lambda f \in \text{FCD} : \top$.

Question What repeated applying of Φ_* and Φ^* to "other" leads to? Particularly, does repeated applying Φ_* and/or Φ^* to the node "other" lead to finite or infinite sets?

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Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>